INCH-POUND

A-A-59254
MAY 13, 1998

COMMERCIAL ITEM DESCRIPTION

ADHESIVE, ANIMAL GLUE, DRY FORM

1. SCOPE: This commercial item description covers the Dry Form (flake, ground, or powdered) of animal glue adhesives for use in woodworking.

2.0 SALIENT CHARACTERISTICS

2.1 Material. The adhesive shall be manufactured from raw materials derived from animals. It shall be homogeneous, free from dirt and foreign matter.

2.2 Requirements. The adhesive shall pass the requirements as specified per the contract including the tests and limits given below.

TABLE 1.

GRADE	J1	J2
Moisture Content percent *Note 1	12	12
Viscosity, millipoises, minimum	83 (8.3 mPa.s)	100 (10 mPa.s)
Jelly Strength	237 to 266	299 to 330
pH *Note 2	6.5	6.5
Foam, millimeters, Maximum	25	25
Odor and keeping quality	Free from any order or decomposition after 48 hours incubation	

Notes:
1. Tolerance ± 3 percent
2. Tolerance ± 1.0 pH unit

Beneficial comments, recommendations, additions, clarifications, etc. and any other data which may improve this document should be addressed to: Commander, U.S. Army TACOM-ARDEC, ATTN: AMSTA-AR-QAW-E, Bldg 12, Picatinny Arsenal, NJ 07806-5000

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2.2.1 Testing. The methods of testing specified, as applicable, shall be followed for each lot (production batch). The lot shall be expressed in pounds (kg). The sample unit for testing shall be one pound (0.46 kg) composite, as applicable, selected in equal portions from samples selected at random throughout the lot. The samples shall be selected in the frequency listed below and thoroughly mixed to form the composite. The composite shall be placed in a clean, dry container with a minimum of air space within and then sealed to protect against moisture. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if the average of determinations on the composite sample fails to meet any of the test requirements specified.

LOT SIZE Pounds (kg)	SAMPLE FREQUENCY
800 or less	2
801 up to in including 22,000	3
22,001 or more	5

2.2.2 Moisture Content. Determine the loss in weight of a 10 g specimen, previously reduced to a size that will pass a No. 4 mesh screen, by drying for 16 hours in a well ventilated oven heated at $105^{o} \pm 3^{o}$ C and using the following calculation:

$$\text{Moisture content, percent} = \frac{\text{(original weight of specimen - wt. of specimen after drying) x 100}}{\text{(original weight of specimen)}}$$

2.2.3 Viscosity. Three separate solutions of the adhesive shall be prepared as follows: Enough adhesive sample shall be taken (calculated from the moisture content previously determined under 2.2.2), to give the equivalent of 13.20 ± 0.05 g of dry adhesive (for example, 15 g if the adhesive contains 12 percent moisture) and placed in a standard wide-mouthed bottle, 150-ml capacity with an inside diameter of 59 ± 2 mm, an outside diameter of 66 ± 1 mm, and an overall height of 85 ± 2 mm. The bottle shall be fitted with a No. 9 rubber stopper. Enough distilled water shall be added to the adhesive to make a total of 106.80 ± 0.05 g of water including that naturally contained in the adhesive (for example 105 g of distilled water if the adhesive has 12 percent moisture content). This will give an eleven percent solution of adhesive. The adhesive and water shall be thoroughly mixed by stirring with a glass rod and then allowed to stand at a temperature of 10 to 15°C for at least 16 hours. The solution shall then be melted by raising the temperature of the solution to 60°C. The time required to bring the solution to 60°C shall not exceed 15 minutes.

The viscosity determination shall be made in a standard jacketed pipette viscometer, or any other suitable viscometer capable of measuring viscosity in millipoise (mPa.s). The instrument must be equipped to hold the adhesive solution at approximately a constant temperature. The temperature of the adhesive solution shall be held at 60 ± 0.2°C and the determination made as soon as the solution has uniformly reached the required temperature. Care should be exercised in filling to see that air is not trapped in the viscometer or in the

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adhesive solution. A stopwatch, graduated in fifths of a second, shall be used to measure the time required for the solution to pass through the orifice of the instrument. The viscosity shall be expressed in millipoisis (mPa.s) and shall be the average of three determinations.

2.2.4 Jelly Strength. The samples from the viscosity test shall be placed in the standard containers described under 2.2.3, and used for the jelly strength determination. The adhesive solutions should be free of air or foam. The solutions shall be allowed to cool in the standard stoppered bottles to approximately 45°C and any water of condensation shall be mixed thoroughly with the solution. The bottles shall then be placed in a chamber or batch controlled at 10° ± 0.1°C for 16 to 18 hours. The determination of jelly-strength of each of the three solutions shall be made on a Bloom gelometer or equivalent apparatus adjusted to give a 4 mm depression and to deliver a load of 40 g per second to the surface of the Jelly. The test must be completed within 2 minutes from the removal of the sample from the cooling chamber. The jelling-strength shall be reported in grams and shall be the average of the three determinations.

2.2.5 pH Value. Portions of the solutions upon which the viscosity and jelly-strength tests were made or other samples of equal concentration shall be used to determine the pH of the adhesive. Determine the pH of solution per ASTM E 70, pH of Aqueous Solutions with Glass Electrode.

2.2.6 Foam. Portions of the samples used in the viscosity and jelly-strength tests or other samples of equal concentration shall be used for a determination of the tendency to foam. Seventy-five ml of the solution at 60°C shall be placed in the standard container described under 2.2.3, and agitated with a stirrer or propeller. The stirrer shall be placed in the lower one-half of the solution and the speed of the stirrer during test shall be between 13,000 and 15,000 rpm. The solution shall be agitated for 1 minute and then allowed to stand at room temperature for 10 minutes, at the end of which there shall be not more than 25 mm of foam.

2.2.7 Odor and Keeping Qualities. A 50 ml or larger portion of the adhesive solution used in previous tests shall be placed in a covered container and held in an incubator at 37 to 38°C for 48 hours. At the end of this time, it shall be free from any odor or decomposition.

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3. QUALITY ASSURANCE PROVISIONS.

The products provided shall conform to the producer's own drawings, specifications, standards and quality assurance practices and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PACKAGING.

The packaging and packing shall be as specified in the contract or order

5. NOTES.

5.1 Intended Use. Animal Adhesive (Dry Form) is intended to be used for edge joint in furniture, etc.

ASTM Methods:
ASTM E 70, pH of Aqueous Solutions with Glass Electrode
ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshocken, PA 19428-2959. The issue of the ASTM test methods in effect on the date of the solicitation shall be used to determine compliance with these requirements.

Custodian:
Army - AR

Preparing Activity:
Army-AR
(Project No. 8040-0556)

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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYYYMMDD)
	A-A-59254	13 MAY 1998

DOCUMENT TITLE
ADHESIVE, ANIMAL GLUE, DRY FORM

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*if applicable*)	7. DATE SUBMITTED (*YYYYMMDD*)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)	
U.S. Army TACOM-ARDEC Standardization Team	(1) Commercial (973) 724-5822	(2) DSN 880-5822
c. ADDRESS (*Include Zip Code*) Attn; AMSTA-AR-QAW-E Picatinny Arsenal, NJ 07806-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Standardization Program Office (DLSC-LM)** **8725 John J. Kingman Road, Suite 2533** **Fort Belvoir, Virginia 22060-6221** **Telephone (703) 767-6888 DSN 427-6888**	

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